

tributed by either donating money or by helping hands-on.

By then I had already redesigned, simulated and revised the plan to maximise efficiency of the dead, depreciated hybrid system, procured necessary electrical items/components and auxiliary material for the project from Kolkata and other markets.

Then, the real work started. We dismantled the dead hybrid system, did some necessary welding and steel fabricating jobs, re-engineered the existing 4.5m (15-ft) tower, installed at Tareybhair fifteen years ago by D.B. Basnet, chief engineer (Retd) for supporting the analogue anemometer to study wind speed. The tower was now left abandoned and came in handy for our installation. An additional 4m (14-ft) tower was created, and both towers were welded and bolted together.

My other motive behind this project was to educate science students of Sadam Senior Secondary School, which was made possible with the school principal Dr Hanna Yonzon's help. Students of class XI provided us with manpower and, in return, were given hands-on experience on how a wind turbine system is installed.

Since then, the whole wind turbine with solar redundancy system has been erected. The system has now been refurbished and renovated, and thoroughly cleaned, greased and oiled. The cable has been laid on the ground and connected with batteries at a distance of 100m, and the whole lighting system in the five un-electrified stalls are connected with a mobile charging facility. Options are available for other battery charging



Bikash Rai with completed wind-solar hybrid system

systems at the self-designed and fabricated control panel, which was the final motive behind the project.

The wind turbine tower was illuminated with two LED flood-lights, sufficient enough to illuminate the surrounding tower areas, and another LED floodlight, designed in-house, was installed at the entry point of the spot. These lights turn on and off automatically or manually in the evening and daylight, respectively.

The layout of the whole system is shown in Fig. 1.

Specifications of the wind turbine with solar redundancy system at Tareybhair are as follows:

System rating. 0.55kW

DC architecture system voltage. 12V

Lighting. LEDs; of the total ten lamps, > 50 watts aggregated

Charging points. Mobile, port-

able batteries charging (optional)

Battery. 120Ah, which is sufficient for 36 hours of back-up, and is enough for the present system. The actual requirement of the whole system is 400Ah for higher full-rated load

Illumination. 4000 lumens aggregated

Average electricity consumption. Presently, it is 289kWh/year, which is very less, and can be upgraded to 1095kWh/year by upgrading the load and battery in the system

The system has been designed in such a way that it just suffices the requirements with a very cost-effective solution.

The cost of the whole system, including dismantling, erection, fabrication, designing, procurement of raw materials and commissioning, came to around ₹ 20,000. It took the team a fortnight to complete the whole project.

Although this is just a small 0.55kW wind-energy generation prototype project, it has led to various awareness programmes to utilise the free-flowing, local wind energy at Tareybhair. Future prospects here are very promising regarding wind-energy based tourism and even small grid-tied wind-power systems.

Aftermath of the prototype

A project proposal of 5kW, along with a street light, at Tareybhair was proposed to member of parliament (Lok Sabha) P.D. Rai of Sikkim, who being an IIT graduate and IIM post-graduate, happily welcome the proposal. He granted an amount of ₹ 1.5 million for the said project. The work has commenced and is expected to be completed by February 2015, with me, Bikash Rai, as principal engineer for the project. ●